Avrile Floro

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EDUCATION

Institut Polytechnique de Paris, France Sep. 2025 – Jul. 2027 (expected)

PhD Track Master in Data & Artificial Intelligence (jointly offered by École Polytechnique and Télécom Paris)

A Master's program with a reinforced research component: individualized supervision by a faculty advisor and integration into a research lab from the first year. Advisor: Prof. Nils Holzenberger. Ranked 1st out of 30 in 2025–2026 (overall average: 19.32/20).

University of Paris 8 Vincennes-Saint-Denis, France Sep. 2022 – Jul. 2025

Bachelor's Degree in Computer Science, with highest honors

University of Artois, France Sep. 2022 – Jul. 2023

Master's Degree in Teaching French as a Foreign, Second, or Specific-Purpose Language, with honors

Paris Dauphine University, France Sep. 2016 – Jul. 2017

Master's Degree in Financial Law, with honors

Master's thesis: *Whistleblower Status and the Scope of Its Protective Provisions*.

Duke University, USA Aug. 2015 – May 2016

Master of Laws (LL.M.) with Certificate in Business Law

Paris-Sud University (now Paris-Saclay University), France Sep. 2012 – Jul. 2015

Bachelor's Degree in Law; University Diploma in Comparative and International Legal Studies, with honors

PUBLICATIONS

[1] *Where Experts Disagree, Models Fail: Detecting Implicit Legal Citations in French Court Decisions*
A. Floro, T. Dhorasoo, S. Pellez, & N. Holzenberger. **[NLLP @ EMNLP 2026]**
In Proceedings of the Natural Legal Language Processing Workshop 2026, pages 156–175, Budapest, Hungary. Association for Computational Linguistics.
Oral presentation. aclanthology.org/2026.nllp-1.15

[2] *Cultural Binding Heads in Language Models* **[BlackboxNLP @ EMNLP 2026]**
A. Floro & L. Benedetto.
In Proceedings of the 9th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP, pages 34–48, Budapest, Hungary. Association for Computational Linguistics.
Poster. aclanthology.org/2026.blackboxnlp-1.3

[3] *A Grammar of French Civil Cassation: A Ground-Level Dataset and Tool* **[JURIX 2026]**
A. Floro, T. Dhorasoo, N. Holzenberger, T. Le Goff, T. Viard, & M. Boritchev.
To appear in Proceedings of the 39th International Conference on Legal Knowledge and Information Systems (JURIX 2026). arXiv:2610.31116

RESEARCH EXPERIENCE

Student Researcher, Interpretability Auditing for Bias Measurement Jun. 2026 – Present

Emory University, USA | Supervisor: Prof. Ali Emami

Research on the reliability of interpretability methods for social-bias auditing: designing evaluation protocols to establish when readouts of a model's internal representations can be trusted for bias measurement, across multiple open-weight LLMs.

Research Intern, Argument Mining on French Supreme Court Rulings May 2026 – Aug. 2026
i3 – Interdisciplinary Institute for Innovation, Télécom Paris (CNRS joint research unit UMR 9217), France
Supervisors: Prof. Tiphaine Viard, Prof. Maria Boritchev, Prof. Thomas Le Goff
First author of *A Grammar of French Civil Cassation* [3], accepted at JURIX 2026. Computational analysis of civil rulings from the French Supreme Court (*Cour de cassation*), using Judilibre open data. Built a large-scale extraction pipeline exploiting the codified rhetorical structure of *cassation* rulings to map each ground of appeal to the court's response, yielding 121K+ argument pairs across 86K decisions and the five civil chambers.

Student Researcher, Mechanistic Interpretability of Cultural Binding Feb. 2026 – Jul. 2026
SAMOVAR, Télécom SudParis, France | Supervisor: Prof. Luca Benedetto
Cultural Binding Heads in Language Models [2]: causal analysis (edge knockout, dose-response α-scaling, generation-time steering) of why LLMs default to equal treatment across cultural groups even when context warrants differentiation. Identified 2–3 mid-layer attention heads mediating cultural binding across 8 LLMs, and showed that models know 3–6$\times$ more than they act upon.

Student Researcher, Implicit Legal Citation Detection Apr. 2025 – Jul. 2026
DIG Team, LTCI, Télécom Paris, France | Supervisor: Prof. Nils Holzenberger
Detecting Implicit Legal Citations in French Court Decisions [1]: built a benchmark of 1,015 passage-article pairs annotated by three legal experts ($\kappa = 0.33$) and showed that annotator disagreement is the strongest predictor of failure across all evaluated models: 68% of false positives fall on the 33% of disputed cases. Designed and implemented the full pipeline: Judilibre data collection, JuriBERT bi-encoder training, adversarial filtering, and supervision of the expert annotation campaign.

HONORS & AWARDS

BlackboxNLP 2026 Travel Grant (Sep. 2026), awarded by the BlackboxNLP organizers: EMNLP 2026 registration fee support (up to $550) for in-person attendance.

IP Paris PhD Track Excellence Scholarship (2025–2026, renewed for 2026–2027), competitive merit-based scholarship from Institut Polytechnique de Paris: €10,000/year cost-of-living grant, plus subsidized tuition.

1st place / 87 teams, *Influencer or Observer: Predicting Social Roles*, Kaggle competition for the Deep Learning course at École Polytechnique, Fall 2025 (see Projects).

PROJECTS

Influencer or Observer: Predicting Social Roles on Twitter – winning entry (88.0% accuracy, 1st of 87 teams) in the Kaggle challenge of the Deep Learning course at École Polytechnique, with Falguny Barua Ema and Saurabh Mishra: stacking ensemble combining LightGBM over 1,373 behavioral features, a fine-tuned CamemBERTa-v2, and a HistGradientBoosting meta-learner. Repository.

Collective Nudging that Scales: How Many Dogs Do I Need to Herd Sheep? – NetLogo agent-based study of how herding effort scales with flock size, with Prof. Ada Diaconescu (LTCI, Télécom Paris); proposes a local sheep-density gradient as a tractable proxy for the flock's centre of mass. Repository.

French Electricity Consumption Forecasting – interactive R Shiny app for rolling forecasts of national electricity demand, with built-in model performance testing. Live demo.

Self-Organizing Map Classifier – Kohonen self-organizing map implemented from scratch in C, used as an unsupervised classifier. Repository.

TEACHING EXPERIENCE

Academic Tutor, Computer Science Bachelor's Program Nov. 2024 – Jul. 2026

University of Paris 8 | 218 hours/year | Appointed for 2024/2025, renewed for 2025/2026

Deliver individualized academic and methodological support across the undergraduate Computer Science curriculum; serve as the primary liaison between faculty and students.

French Teacher Aug. 2020 – Aug. 2026

The French Class, San Francisco, USA

Six years of one-on-one and group instruction across all CEFR levels (A1 to C2); design personalized curricula and weekly teaching materials tailored to each learner; built an automated note-organization tool to streamline lesson follow-up.

WORK EXPERIENCE

Associate Founder Apr. 2019 – May 2020

Bonjour SAS, Bogotá, Colombia

Co-launched a social enterprise selling fair-trade cocoa products sourced from Colombian coca-to-cacao crop substitution programs; handled business planning, digital strategy, and retail partnerships.

Divemaster (PADI) Mar. 2018 – Jul. 2018

Next Level Scuba, Indonesia

Banking Risk & Compliance Consultant (Apprenticeship) Sep. 2016 – Jul. 2017

PricewaterhouseCoopers (PwC), Neuilly-sur-Seine, France

Audited the compliance procedures of global credit institutions; drafted and revised procedures in anticipation of legislative changes; implemented recommendations of the ACPR (French banking and insurance supervisor), governance, and industry best practices.

TECHNICAL SKILLS & LANGUAGES

Programming: Python (proficient), R, C, Java, Lisp
ML / NLP stack: PyTorch, Hugging Face Transformers, scikit-learn, pandas, NumPy
Other tools: R Shiny, NetLogo, Git, LaTeX
Languages: French (native), English (C2, IELTS 8.5/9, Aug. 2026), Spanish (B2)